INTRODUCTION TO DEEP LEARNING

Course number: 00240332

Lecture 13: Generative Adversarial Networks

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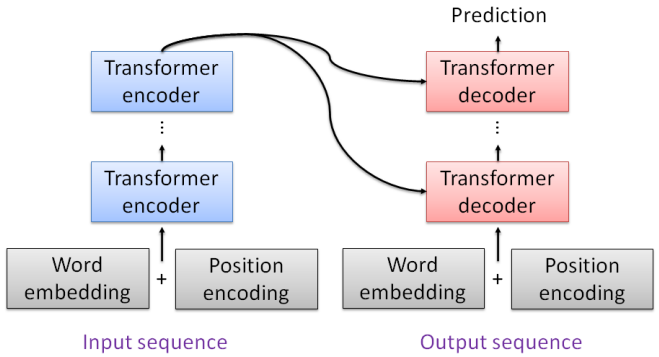
Last lecture review

I. Self-attention

$$\boldsymbol{a}_t = \sum_{i=1}^{N} \text{softmax}\left(e_i^t(\boldsymbol{k}_i, \boldsymbol{q}_t)\right) \boldsymbol{v}_i$$

key | query | value

II. Transformer



Self-attention + Cross-attention

Encoder + Decoder

III. BERT

Transformer Encoder

III. GPT

Transformer Decoder

Two types of machine learning models

Discriminative models

- Conditional distribution P(y|x)
- Supervised learning: assign correct labels to data
- Examples
  - Naïve Bayes
  - Hidden Markov model
  - Support vector machine
  - Multilayer perceptron
  - Convolutional neural network

Some models can be used as discriminative models or generative models!

Generative models

- Joint distribution P(x,y)
- Unsupervised learning: discover the hidden structure in the data
- Examples
  - Restricted Boltzmann Machine
  - Deep Belief Network
  - Deep Boltzmann Machine
  - Denoising Autoencoders
  - GPT
  - Gerative adversarial nets

Outline

I. The basic GAN
II. GANs for high-quality results
III. GANs for controllable generation
IV. Summary

Motivation

Target: design a generator (G) to generate images which are *very much like* the images in the training set!

How to measure the similarity?

Min KL-divergence between $P_{\text{data}}(\mathbf{x})$ and $P_{\text{gen}}(\mathbf{x})$ → Max likelihood $P_{\text{gen}}(\mathbf{x})$ → Predefine a form for $P_{\text{gen}}(\mathbf{x})$

https://wiseodd.github.io/techblog/2017/01/26/kl-mle/

Strong assumption!

Who will judge the quality of generated data? Human!

Train a neural network to do this judgement?

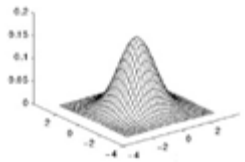
GAN: Generative Adversarial Networks

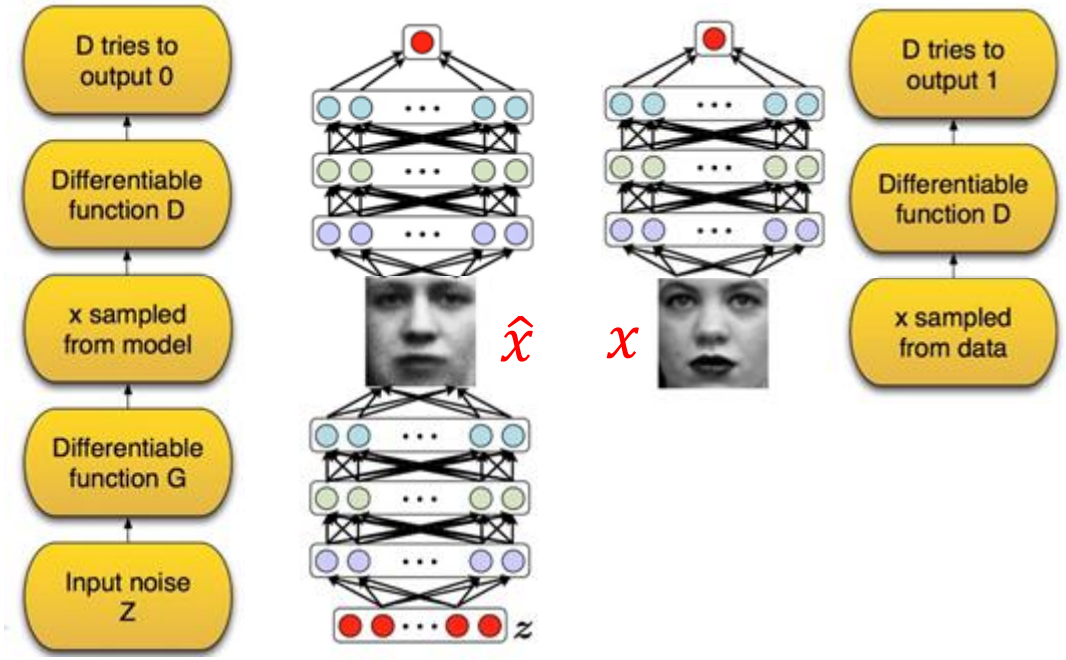
Goodfellow et al, NIPS 2014

Nonlinear mappings,
e.g., MLP and CNN

Discriminator →

Generator →





- D's duty: For $\mathbf{x} \sim P_{\text{data}}$, D should be high; for $\hat{\mathbf{x}} \sim P_{\text{gen}}$, D should be low
- G's duty: Make $D(\hat{\boldsymbol{x}})$ as high as possible where $\hat{\boldsymbol{x}} = G(\boldsymbol{z})$

Minimax two-player game

1. For $\mathbf{x} \sim P_{\mathrm{d}}$, D should be high; for $\hat{\mathbf{x}} \sim P_{\mathrm{g}}$, D should be low

- Let $D \in (0,1)$, the above statement is equivalent to
  - $\mathbb{E}_{\mathbf{x} \sim p_{\mathrm{d}}(\boldsymbol{x})}[\log D(\boldsymbol{x})]$ should be high;
  - $\mathbb{E}_{\mathbf{x} \sim p_{\mathrm{g}}(\boldsymbol{x})}[\log D(\hat{\mathbf{x}})] = \mathbb{E}_{\mathbf{z} \sim p_z(\boldsymbol{z})}\left[\log D\left(G(\boldsymbol{z})\right)\right]$ should be low

$\hat{\mathbf{x}} = G(\boldsymbol{z})$

$\mathbb{E}_{\mathbf{z} \sim p_z(\boldsymbol{z})}\left[\log\left(1 - D\left(G(\boldsymbol{z})\right)\right)\right]$ should be high

- To encourage this result we can

$$\max_D V(D,G) = \mathbb{E}_{\mathbf{x} \sim p_d(\boldsymbol{x})}[\log D(\boldsymbol{x})] + \mathbb{E}_{\mathbf{z} \sim p_z(\boldsymbol{z})}\left[\log\left(1 - D\left(G(\boldsymbol{z})\right)\right)\right]$$

Minimax two-player game

1. For $\mathbf{x} \sim P_{\mathrm{d}}$, D should be high; for $\hat{\mathbf{x}} \sim P_{\mathrm{g}}$, D should be low

$$\max_D V(D,G) = \mathbb{E}_{\mathbf{x}\sim p_{\mathrm{d}}(\boldsymbol{x})}[\log D(\boldsymbol{x})] + \mathbb{E}_{\mathbf{z}\sim p_z(\boldsymbol{z})}\left[\log\left(1 - D\left(G(\boldsymbol{z})\right)\right)\right]$$

2. Make $D(\hat{\boldsymbol{x}})$ as high as possible, where $\hat{\boldsymbol{x}} = G(\boldsymbol{z})$, to fool D

$$\min_G V(D,G) = \mathbb{E}_{\mathbf{z}\sim p_z(\boldsymbol{z})}\left[\log\left(1 - D\left(G(\boldsymbol{z})\right)\right)\right]$$

In summary

$$\min_G \max_D V(D,G) = \mathbb{E}_{\mathbf{x}\sim p_{\mathrm{d}}(\boldsymbol{x})}[\log D(\boldsymbol{x})] + \mathbb{E}_{\mathbf{z}\sim p_z(\boldsymbol{z})}\left[\log\left(1 - D\left(G(\boldsymbol{z})\right)\right)\right]$$

Nothing to do with G

Algorithm: train D and G alternatively

单选题 1分

设置

During maximizing $V(D,G)$ w.r.t. D, we need to input both real data $\mathbf{x}$ and generated data $\hat{\mathbf{x}}$

A Yes

B No

提交

单选题 1分

设置

During minimizing $V(D,G)$ w.r.t. G, we need to input both real data $\mathbf{x}$ and generated data $\hat{\mathbf{x}}$

A Yes

B No

提交

Training algorithm

Algorithm 1 Minibatch stochastic gradient descent training of generative adversarial nets. The number of steps to apply to the discriminator, k, is a hyperparameter. We used $k = 1$, the least expensive option, in our experiments.

for number of training iterations **do**

  for k steps **do**

  - Sample minibatch of m noise samples $\{\boldsymbol{z}^{(1)}, \ldots, \boldsymbol{z}^{(m)}\}$ from noise prior $p_g(\boldsymbol{z})$.
  - Sample minibatch of m examples $\{\boldsymbol{x}^{(1)}, \ldots, \boldsymbol{x}^{(m)}\}$ from data generating distribution $p_{\text{data}}(\boldsymbol{x})$.
  - Update the discriminator by ascending its stochastic gradient:

$$\nabla_{\theta_d} \frac{1}{m} \sum_{i=1}^{m} \left[\log D\left(\boldsymbol{x}^{(i)}\right) + \log\left(1 - D\left(G\left(\boldsymbol{z}^{(i)}\right)\right)\right) \right].$$

  end for

  - Sample minibatch of m noise samples $\{\boldsymbol{z}^{(1)}, \ldots, \boldsymbol{z}^{(m)}\}$ from noise prior $p_g(\boldsymbol{z})$.
  - Update the generator by descending its stochastic gradient:

$$\nabla_{\theta_g} \frac{1}{m} \sum_{i=1}^{m} \log\left(1 - D\left(G\left(\boldsymbol{z}^{(i)}\right)\right)\right).$$

end for

The gradient-based updates can use any standard gradient-based learning rule. We used momentum in our experiments.

单选题 1分

设置

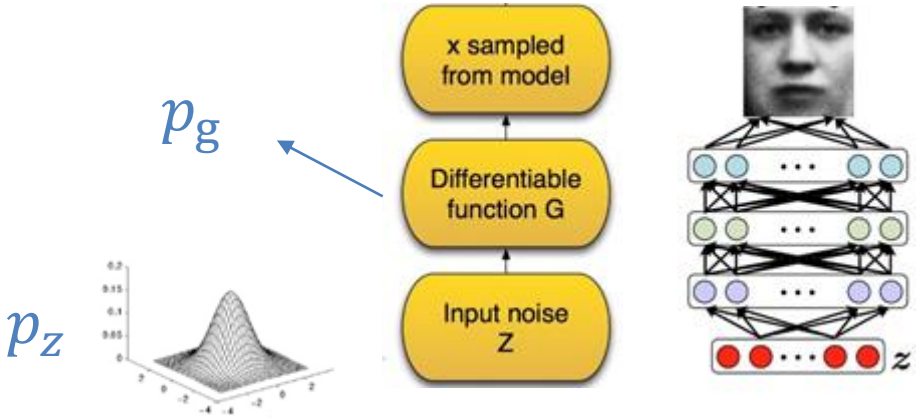
Can we judge if GAN has converged or not by watching the value of $V(D,G) = \mathbb{E}_{\mathbf{x}\sim p_{\mathrm{d}}(\boldsymbol{x})}[\log D(\boldsymbol{x})] + \mathbb{E}_{\mathbf{z}\sim p_z(\boldsymbol{z})}\left[\log\left(1 - D\left(G(\boldsymbol{z})\right)\right)\right]$?

A Yes

B No

提交

Theoretical Results



- The generator G implicitly defines a probability distribution p_g as the distribution of the samples $G(z)$ obtained when $z \sim p_z$

Theorem: the algorithm converges to $p_g = p_d$

Experiments



a) MNIST



b) TFD



c)

CIFAR-10 (fully connected model)



d)

CIFAR-10 (convolutional discriminator and “deconvolutional” generator)

Influence



Ian Goodfellow



[Heroes of Deep Learning: Andrew Ng interviews Ian Goodfellow - YouTube]

Generative adversarial networks

I Goodfellow, J Pouget-Abadie, M Mirza, B Xu... - Communications of the ..., 2020 - dl.acm.org

Generative adversarial networks are a kind of artificial intelligence algorithm designed to solve the **generative** modeling problem. The goal of a **generative** model is to study a collection ..

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Till 2022/12/16

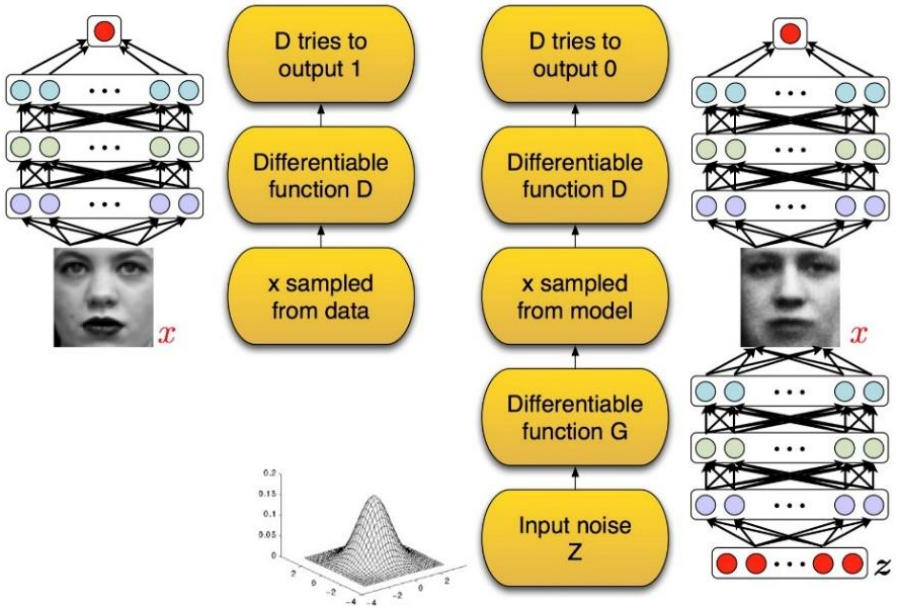
Properties of the GANs

- Unlike many other generative models (e.g. RBM), after training, it doesn't need iterative sampling to do generation

- Most GANs don't have a loss function as the indication of convergence
  - But see the Wasserstein GAN (Arjovsky, ICML 2017)

Summary of Part I





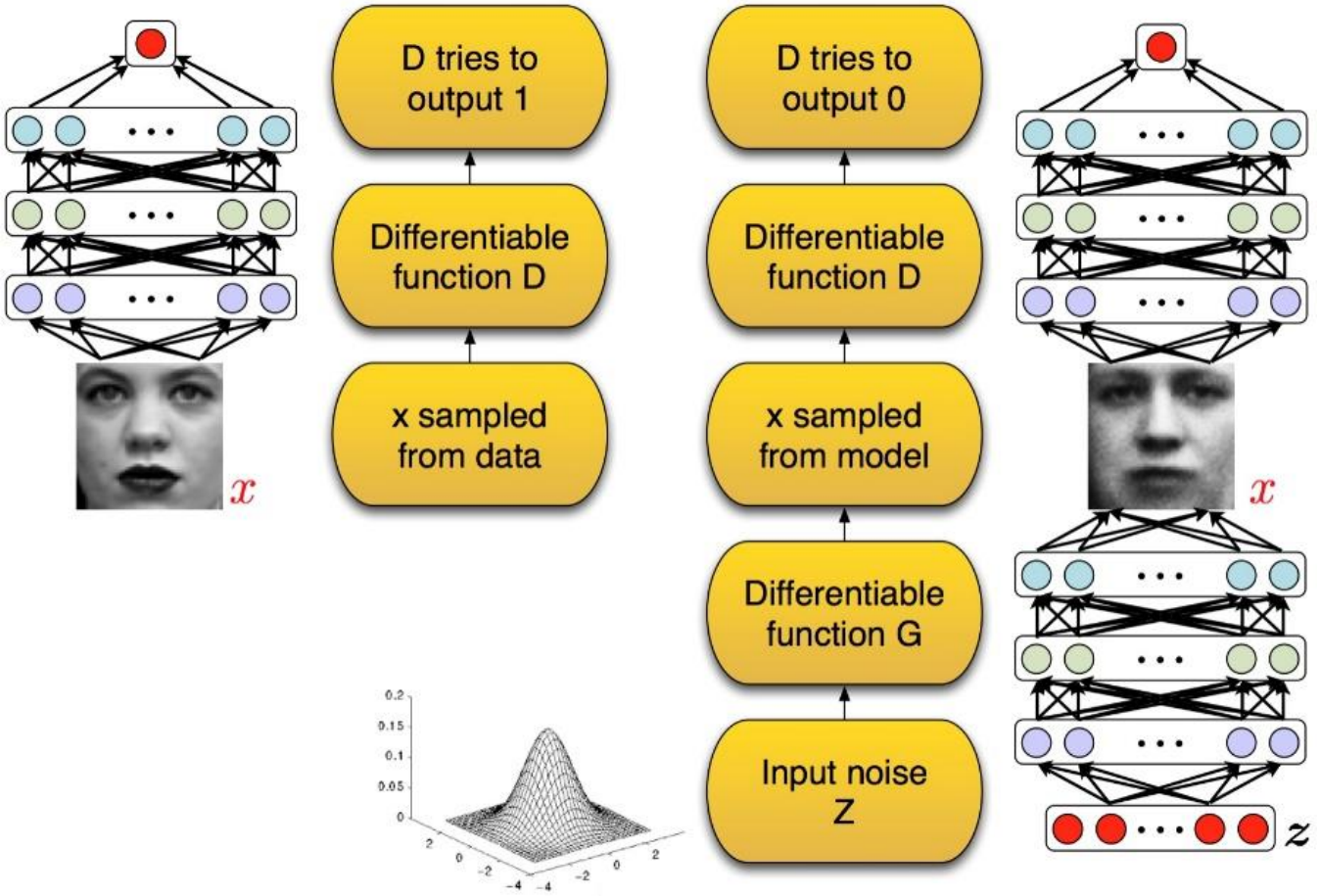
$$\min_G \max_D V(D,G) = \mathbb{E}_{\mathbf{x}\sim p_{\mathrm{d}}(\boldsymbol{x})}[\log D(\boldsymbol{x})] + \mathbb{E}_{\mathbf{z}\sim p_z(\boldsymbol{z})}\left[\log\left(1 - D\big(G(\boldsymbol{z})\big)\right)\right]$$

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II. GANs for high-quality results
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Why not CNN for D and G?

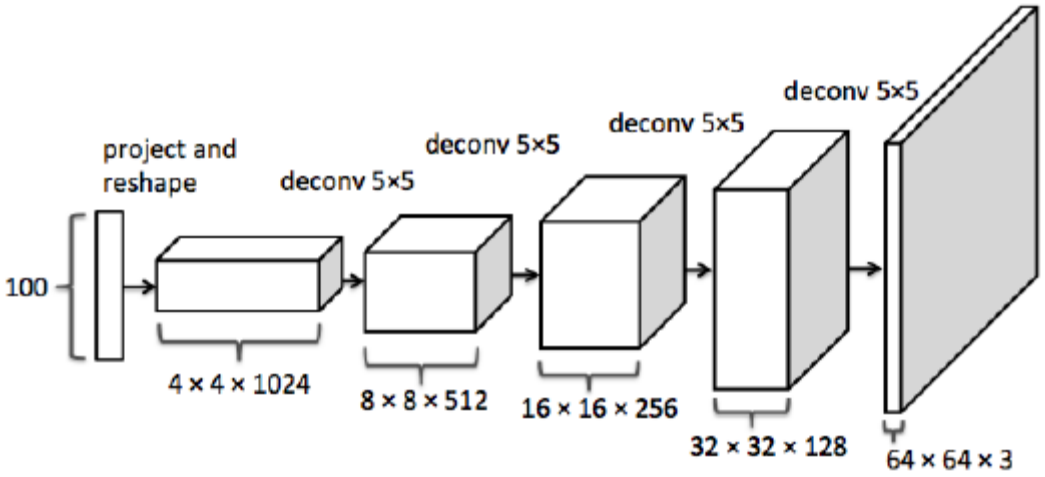
Goodfellow et al, NIPS 2014



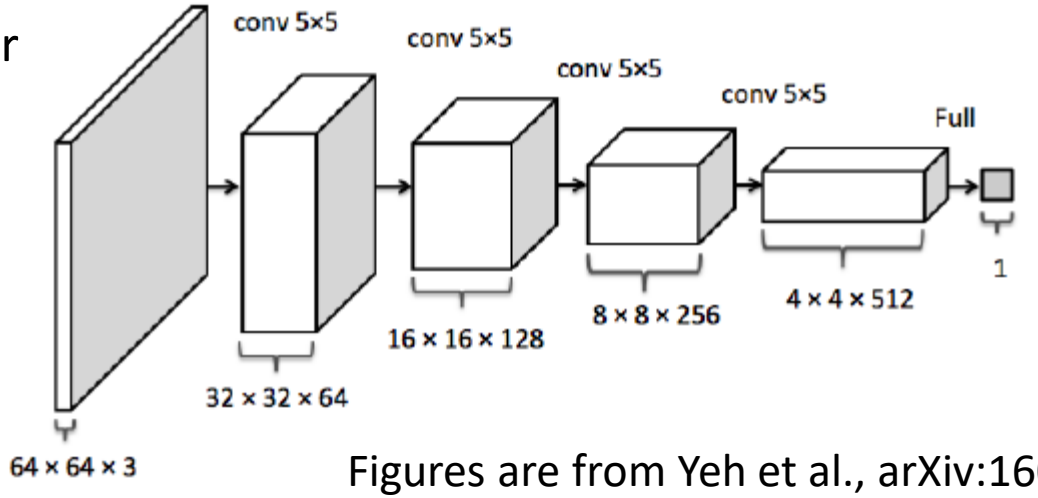
Deep convolutional GAN (DCGAN)

Radford et al., 2016

Generator



Discriminator



Figures are from Yeh et al., arXiv:1607.07539, 2016

Main tricks

- Replace any pooling layers with strided convolutions (discriminator) and fractional-strided convolutions (generator)
- Use batchnorm in both the generator and the discriminator.
- Remove fully connected hidden layers for deeper architectures
- Use ReLU activation in generator for all layers except for the output, which uses Tanh
- Use LeakyReLU activation in the discriminator for all layers

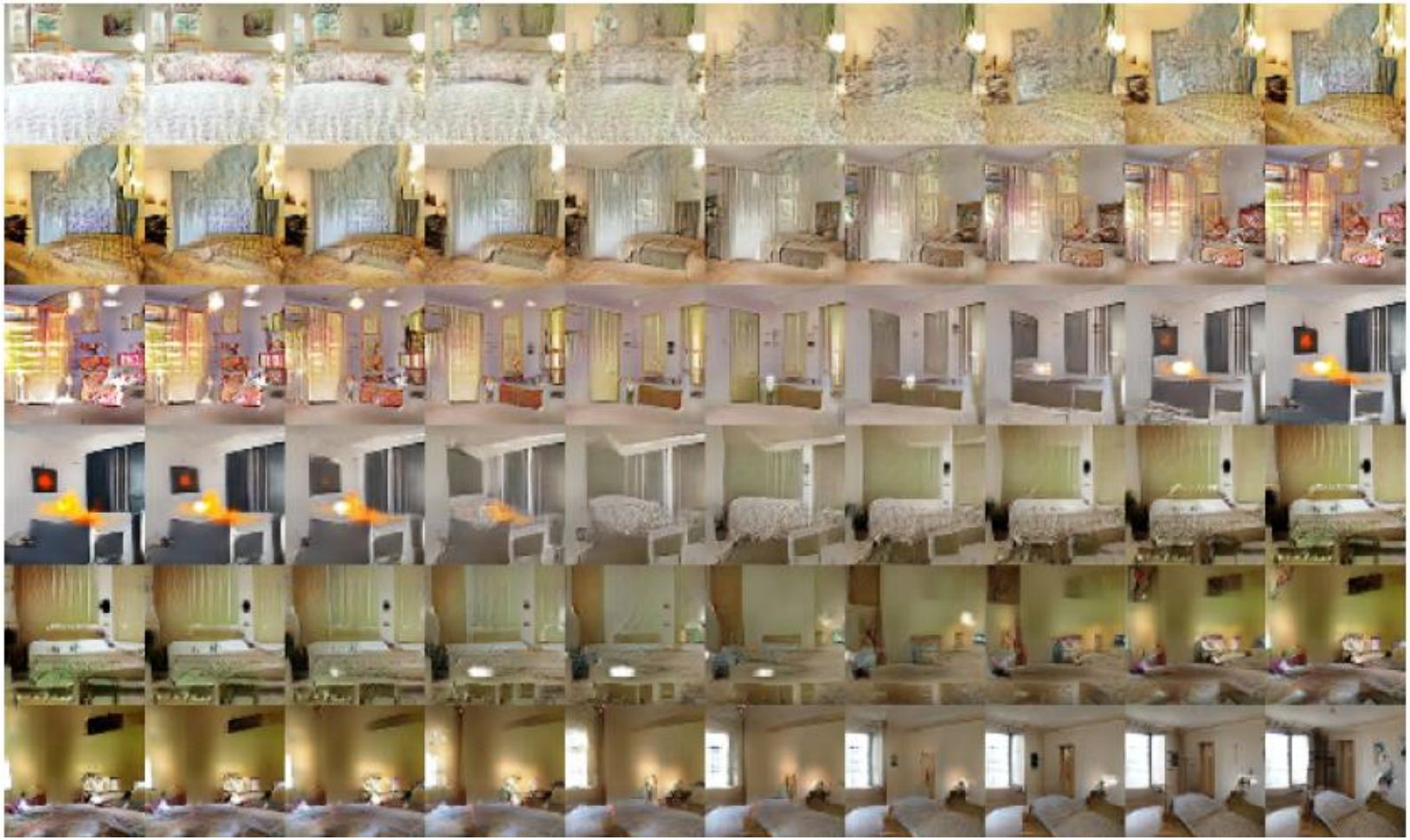
Experiments on LSUN

64*64



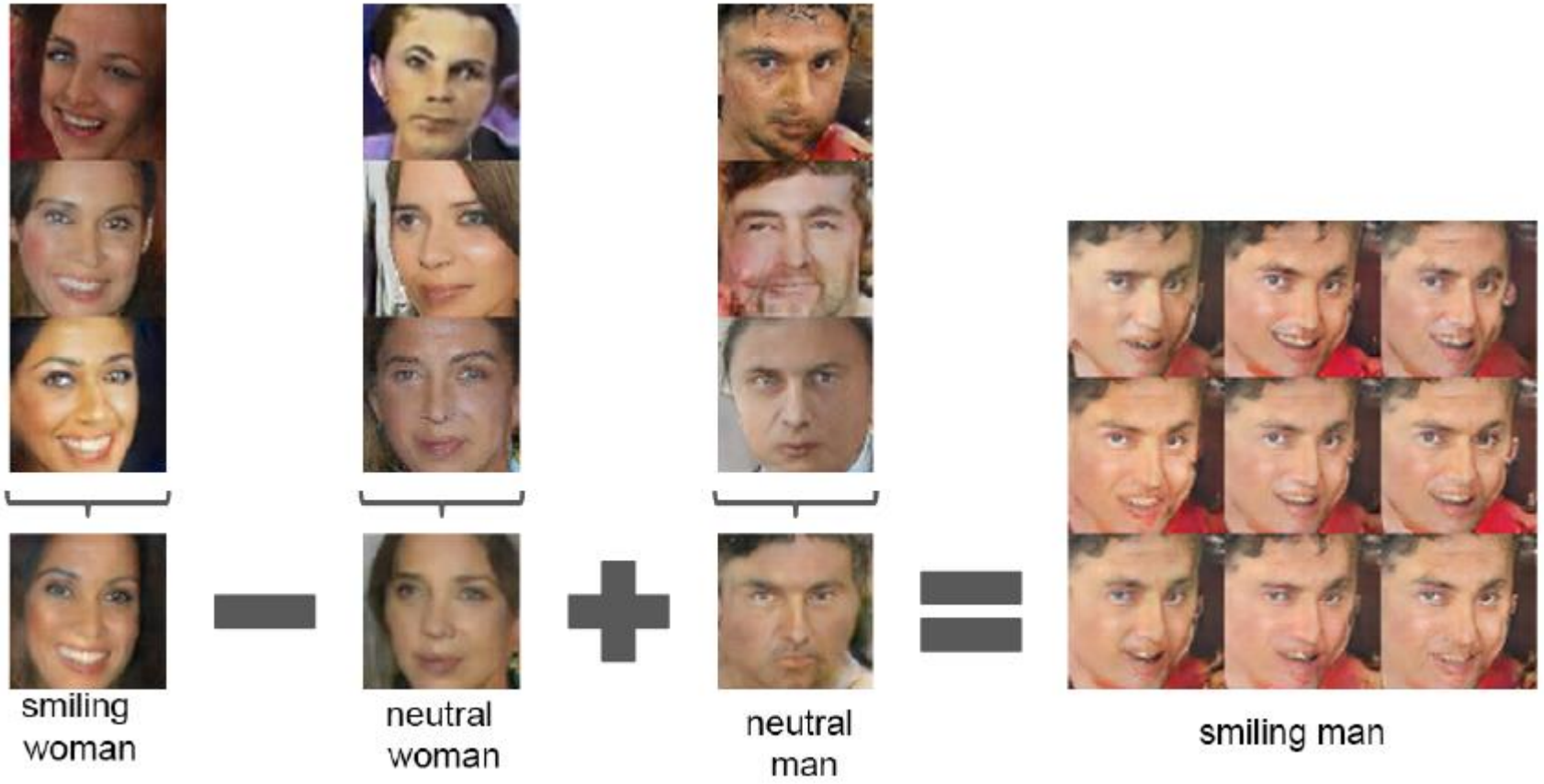
Generated bedrooms after five epochs of training

Walking in the latent space

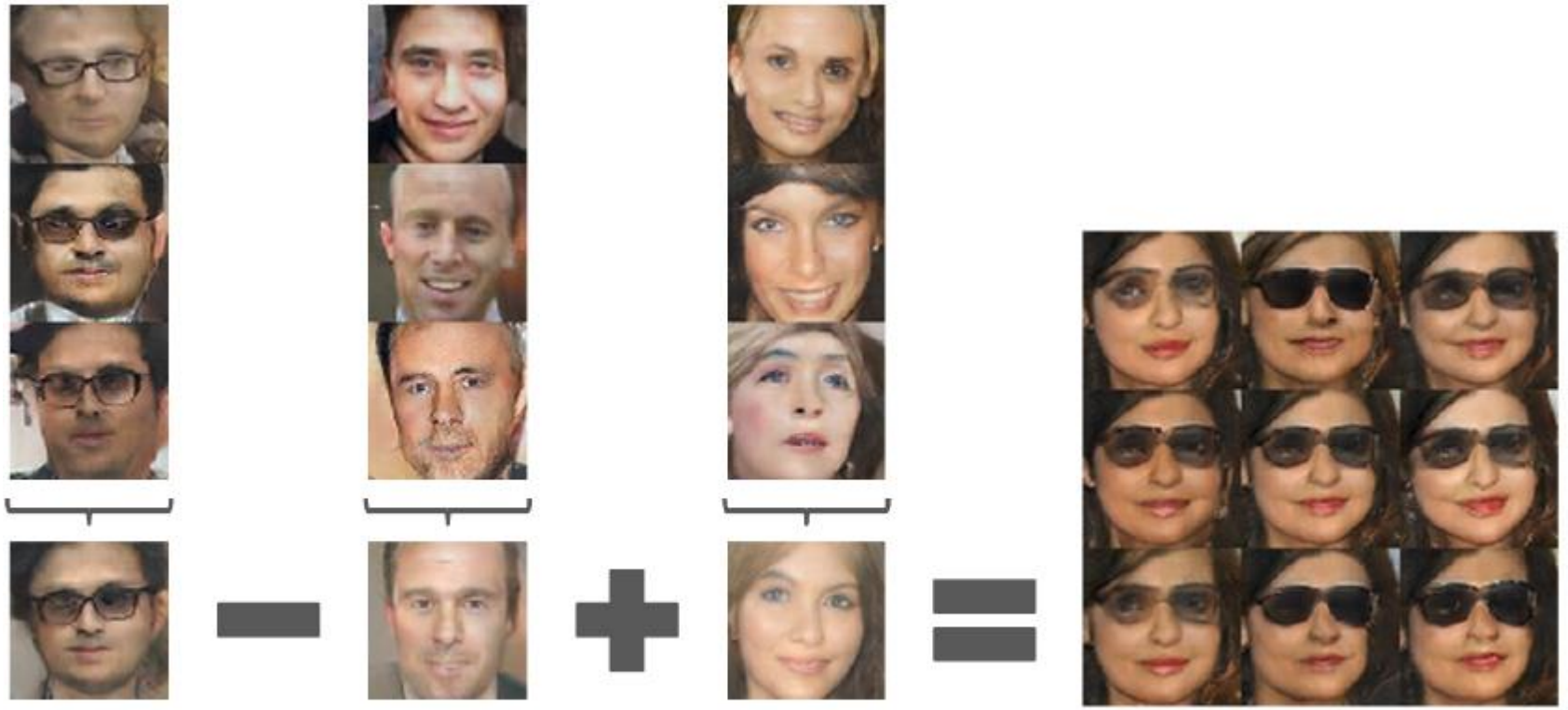


Interpolation between a series of 9 random points in Z

Vector arithmetic on face samples



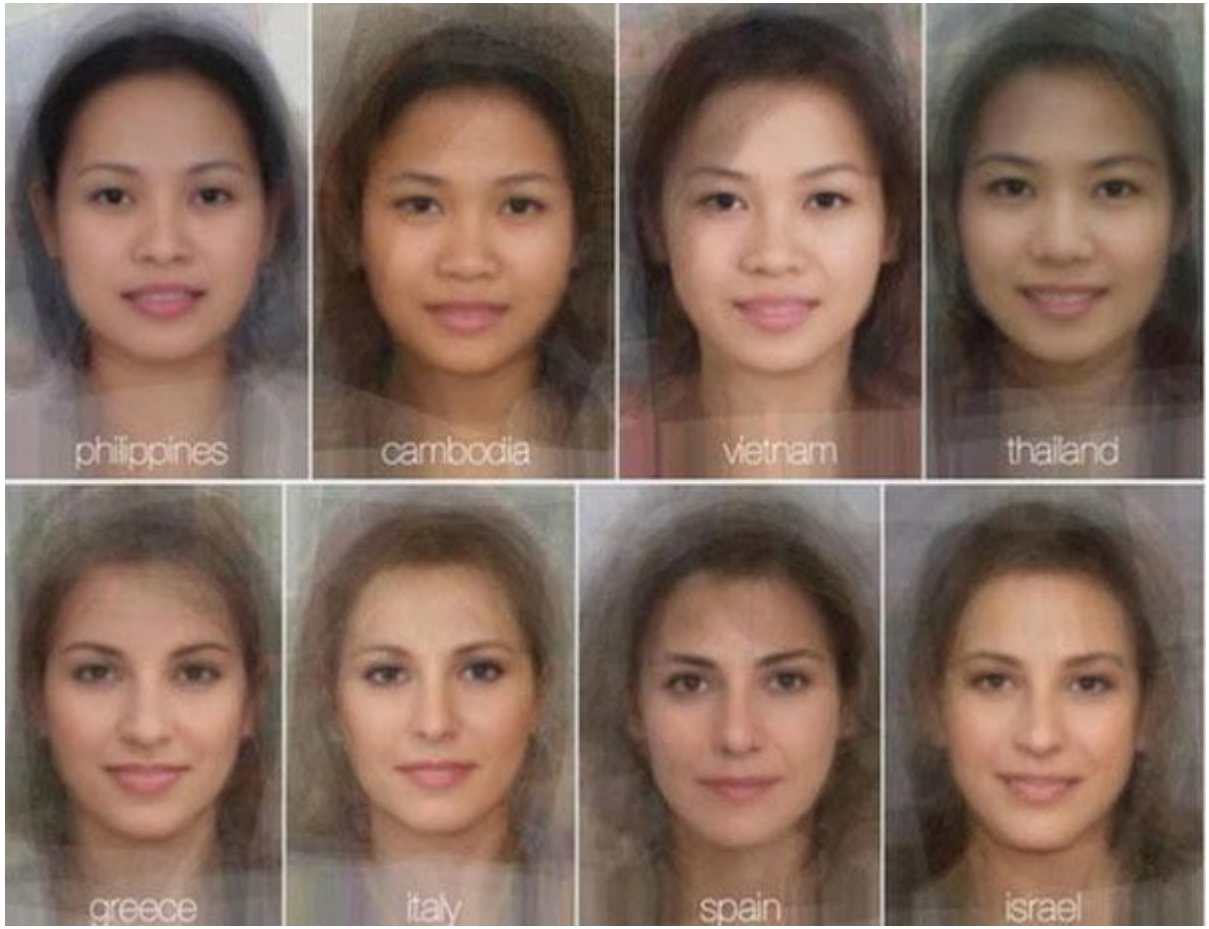
Vector arithmetic on face samples



Vector arithmetic in the pixel space



Average face of different countries



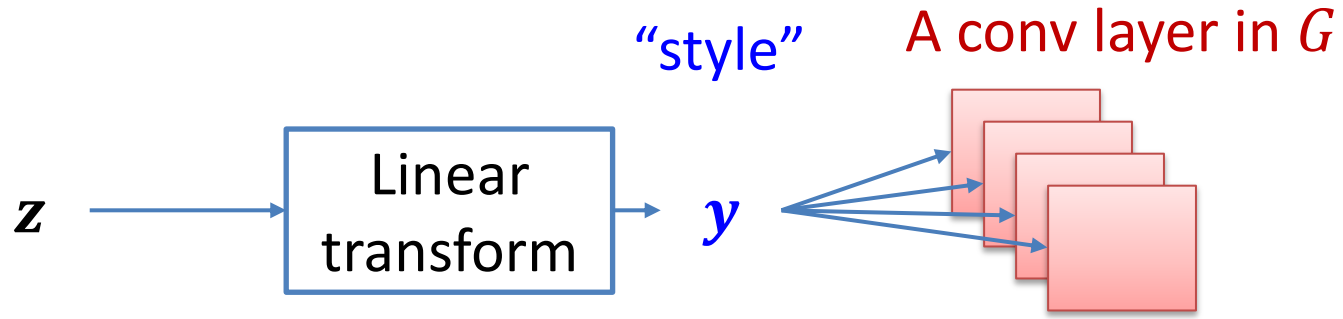
Do you have a better idea?

https://www.artfido.com/this-is-what-the-average-person-looks-like-in-each-country/

A more interpretive generator

Generators operate as black boxes, and the image synthesis process is poorly understood

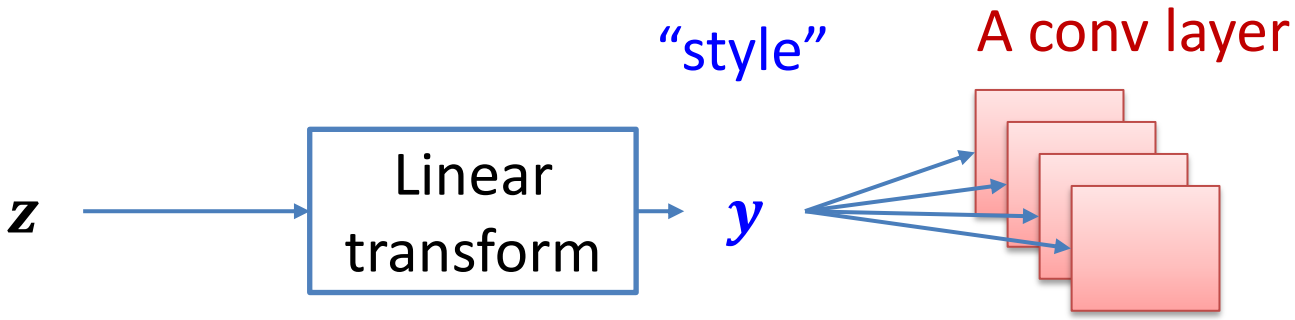
A possible solution



Diff components of y control diff feature maps

(Every feature map may corresponds to a feature of the image, e.g., hue or pose)

How to control the features?



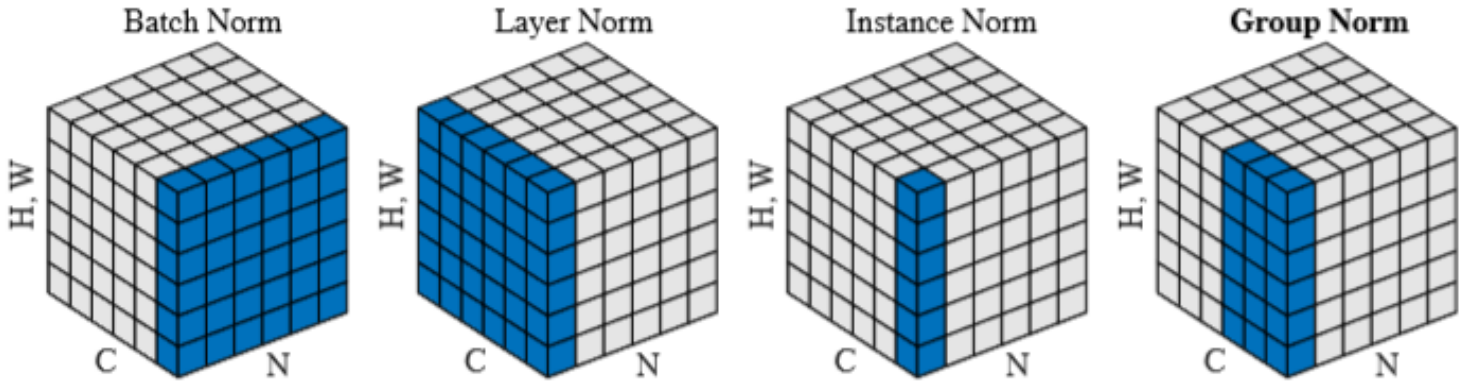
Let $\boldsymbol{y} = (\boldsymbol{y}_s, \boldsymbol{y}_b)$, where $\boldsymbol{y}_s, \boldsymbol{y}_b \in R^M$.
For i-th feature map $\boldsymbol{x}_i$, where $i = 1, \ldots, M$

$$\text{Output} = y_{s,i} \frac{\boldsymbol{x}_i - \mu(\boldsymbol{x}_i)}{\sigma(\boldsymbol{x}_i)} + y_{b,i}$$

where μ and σ denote mean and std

This operation is called *AdaIN*, which was proposed for style transformation (Huang and Belongie, 2017)

Recap: normalization techniques



Wu, He, arXiv:1803.08494v3

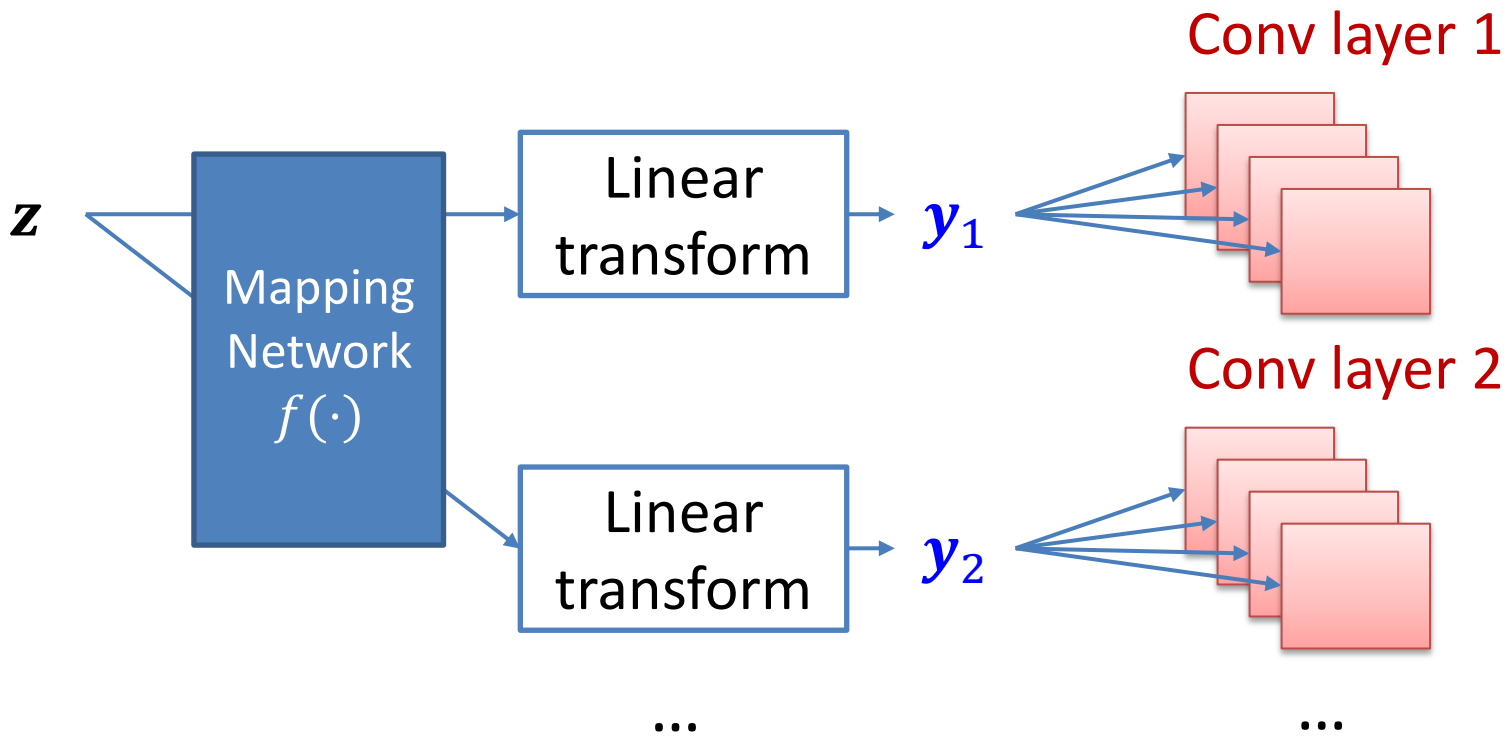
All of them share the same linear transformation form, but do normalization over different dimensions

For i-th feature map $\boldsymbol{x}_i$, where $i = 1, \ldots, M$

$$\text{Output} = y_{s,i} \frac{\boldsymbol{x}_i - \mu(\boldsymbol{x}_i)}{\sigma(\boldsymbol{x}_i)} + y_{b,i}$$

What type of normalization is this?

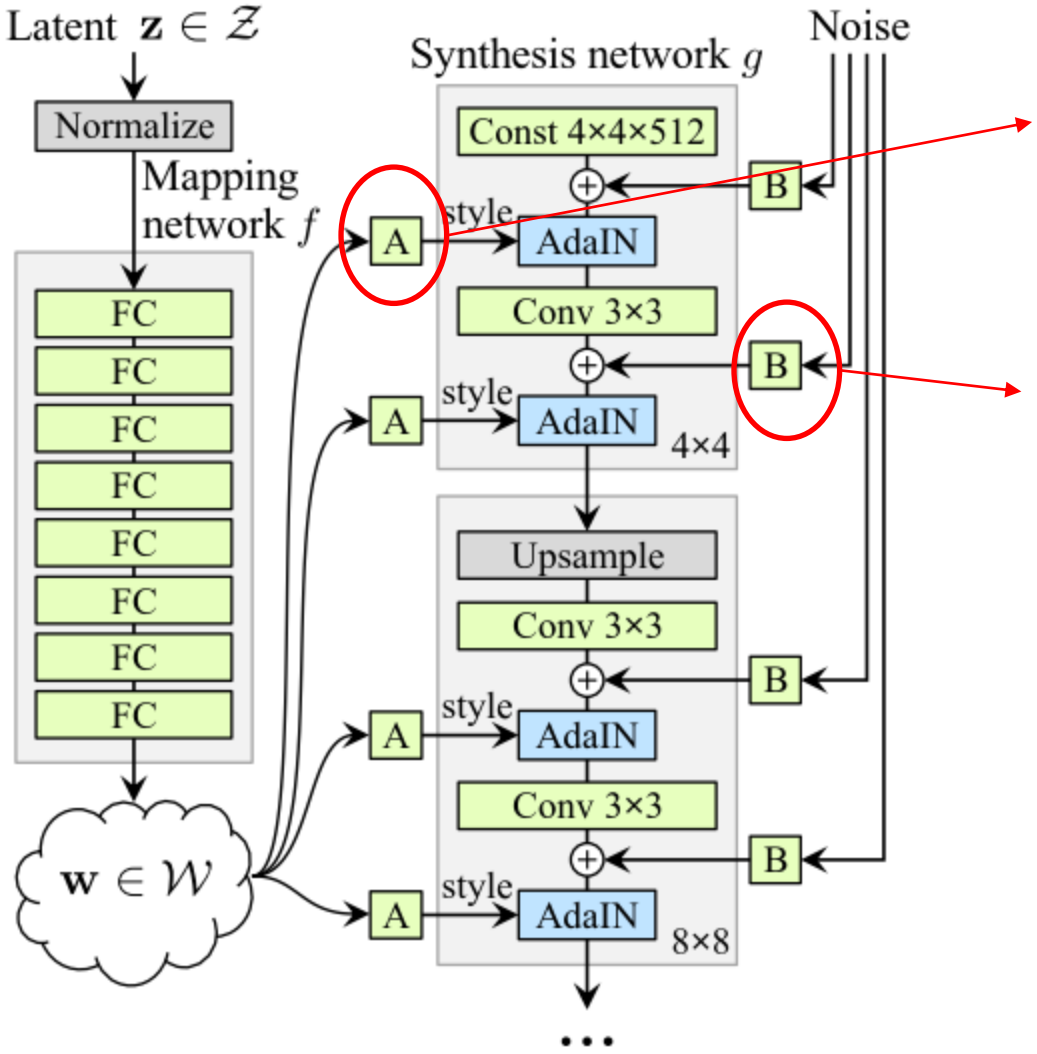
How to control the features?



- AdaIN can be applied to multiple conv layers
- The input z can be first transformed nonlinearly before linearly (see (Karras et al., CVPR 2019) for the reason)

StyleGAN

Karras et al., CVPR 2019



Linear transform

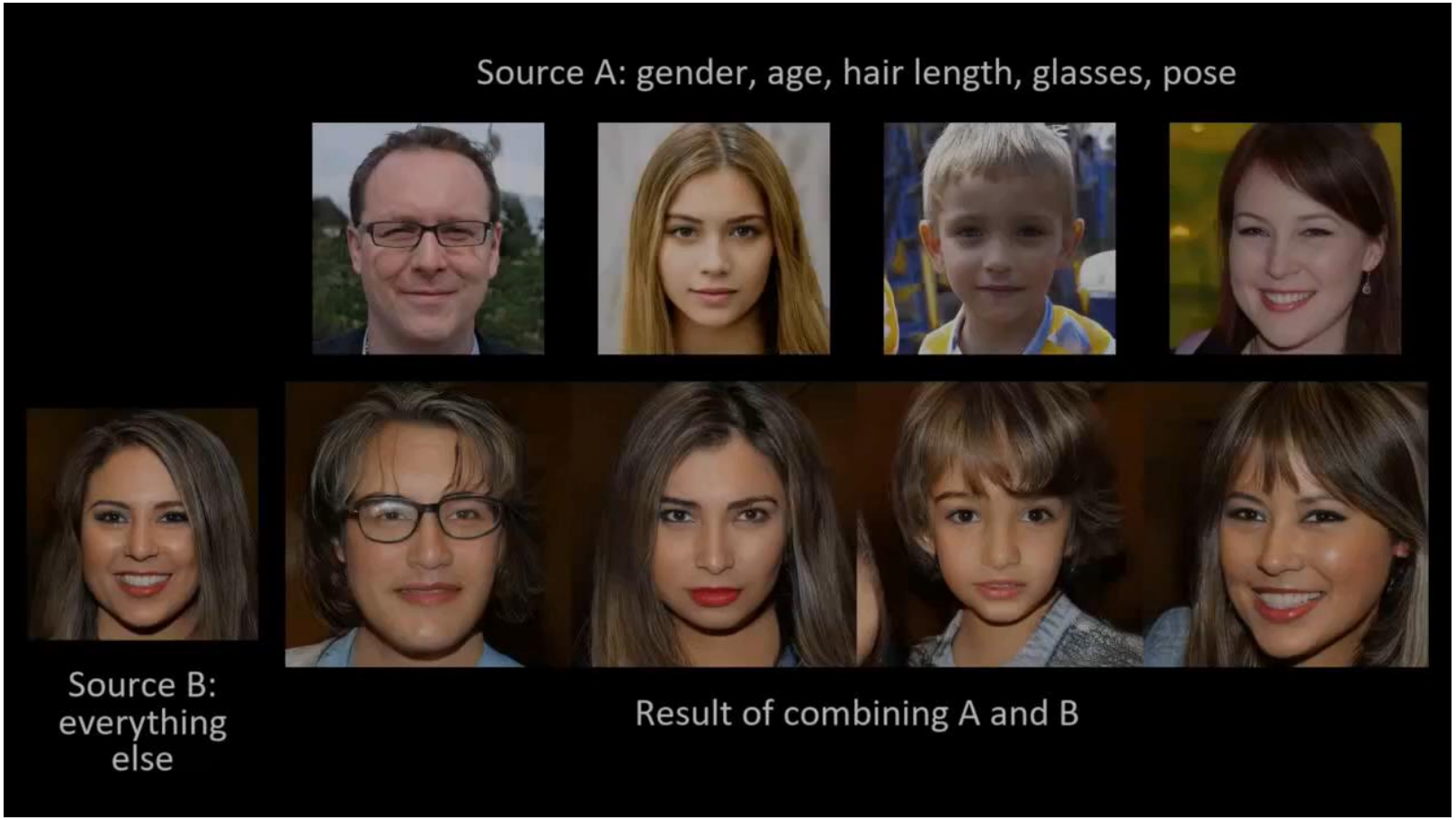
Add noise

Sample results



1024*1024

Mixing two latent codes

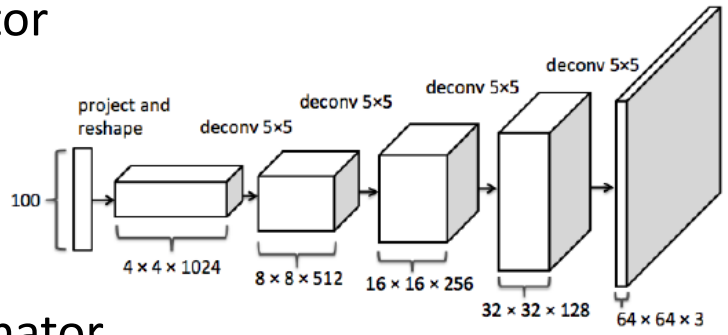


Summary of Part II

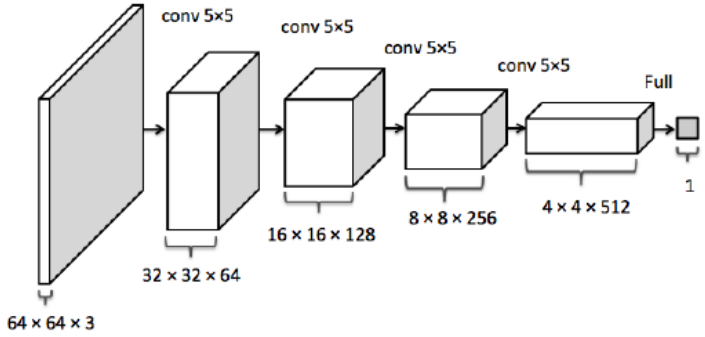
DCGAN

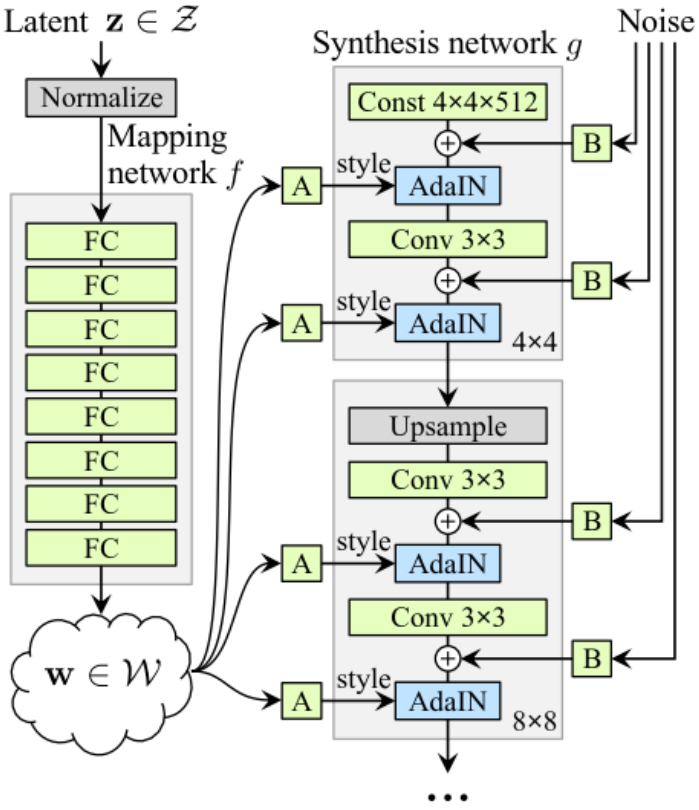
StyleGAN

Generator



Discriminator





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Conditional GAN

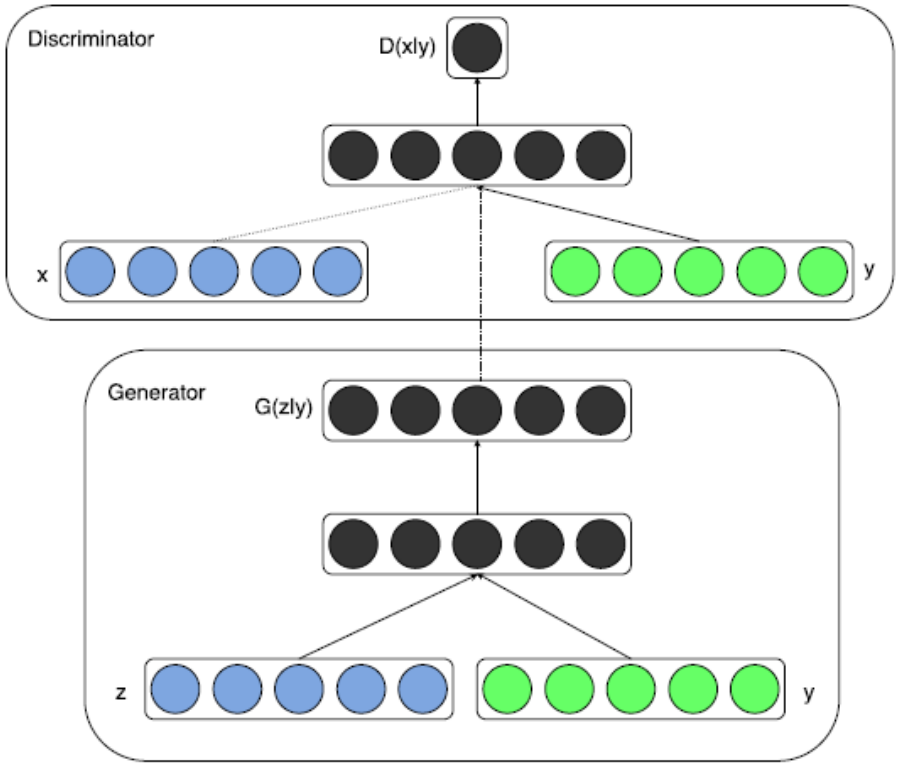
Mirza, Osindero, 2014

- GAN has too much freedom for generation
- A natural method to remedy this problem is adding some constraints
- Let's introduce conditional variable $\boldsymbol{y}$ to both D and G, where $\boldsymbol{y}$ encodes additional information
  - e.g., partial info for image restoration, other modality, label

Conditional GAN

- The objective function of a two-player minimax game

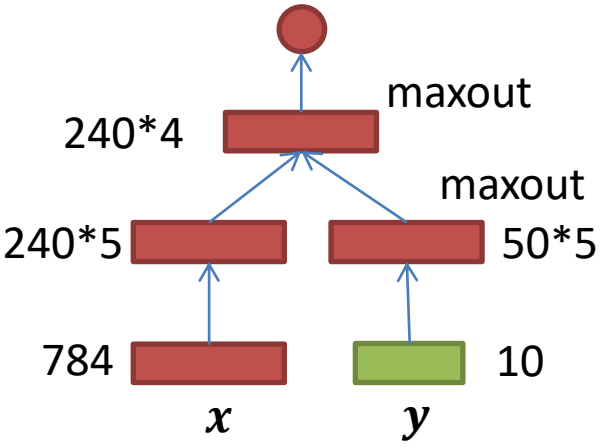
$$\min_G \max_D V(D,G) = \mathbb{E}_{\mathbf{x}\sim p_d(\boldsymbol{x})}[\log D(\boldsymbol{x}|\boldsymbol{y})] + \mathbb{E}_{\mathbf{z}\sim p_z(\boldsymbol{z})}\left[\log\left(1 - D\left(G(\boldsymbol{z}|\boldsymbol{y})\right)\right)\right]$$



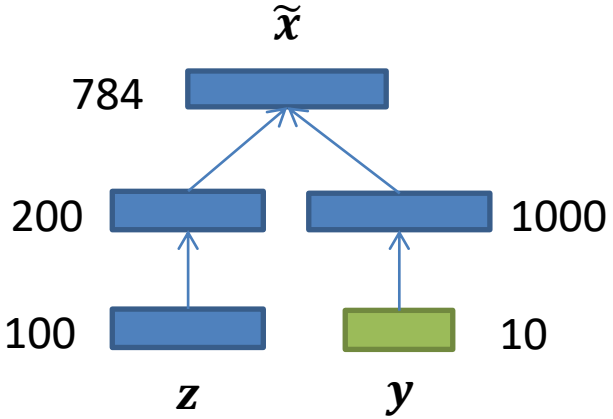
Experiments on MNIST

- Let $\boldsymbol{x}$ be image pixels and $\boldsymbol{y}$ be the one-hot representation of image class

Discriminator network



Generator network



Experiments on MNIST

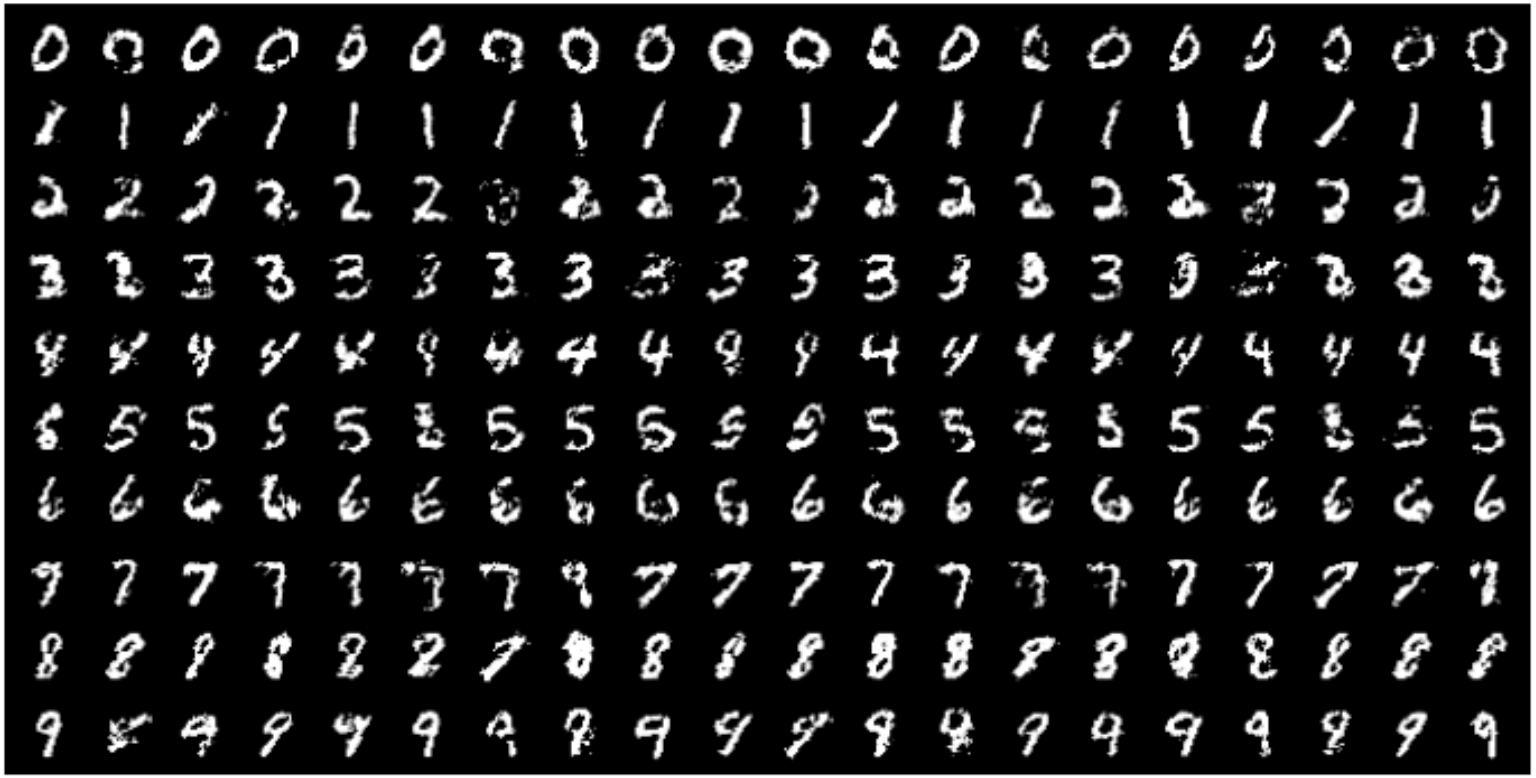


Figure 2: Generated MNIST digits, each row conditioned on one label

Experiments on image tagging

- Let $\boldsymbol{x}$ be tag representation and $\boldsymbol{y}$ be image representation, both of which are pretrained and fixed

	User tags + annotations	Generated tags
	montanha, trem, inverno, frio, people, male, plant life, tree, structures, transport, car	taxi, passenger, line, transportation, railway station, passengers, railways, signals, rail, rails
	food, raspberry, delicious, homemade	chicken, fattening, cooked, peanut, cream, cookie, house made, bread, biscuit, bakes
	water, river	creek, lake, along, near, river, rocky, treeline, valley, woods, waters

单选题 1分

设置

Which task does not need conditional GANs?

A Generate a set of images similar to ImageNet

B Generate image tags given an image

C Generate an image given image tags

D Generate a cartoon image given a real image

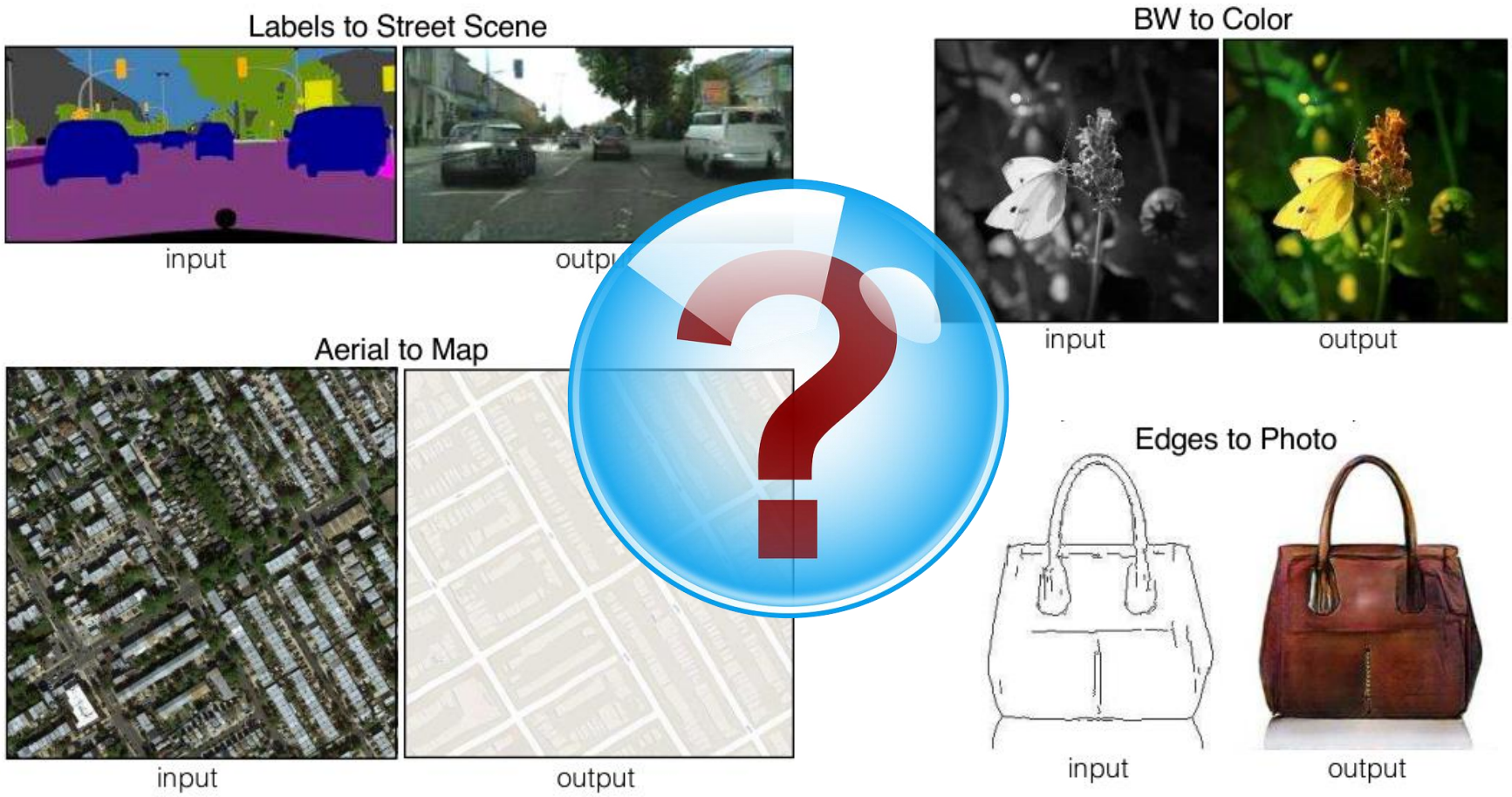
提交

Discussion

- Can you do anything else with the conditional GAN?

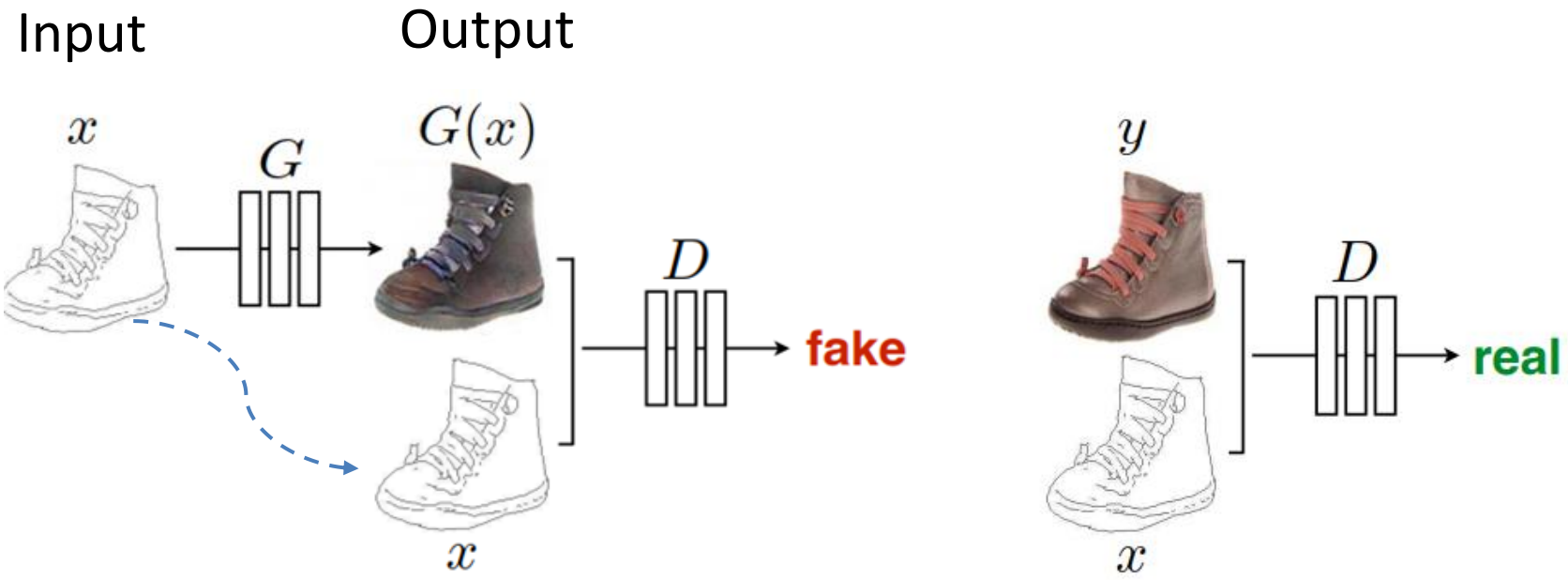
Image-to-image translation

Isola et al., CVPR 2017



The Pix2Pix GAN

Isola et al., CVPR 2017



$$\mathcal{L}_{cGAN}(G,D) = \mathbb{E}_{x,y}[\log D(x,y)] + \mathbb{E}_{x,z}[\log(1 - D(x, G(x,z))]$$

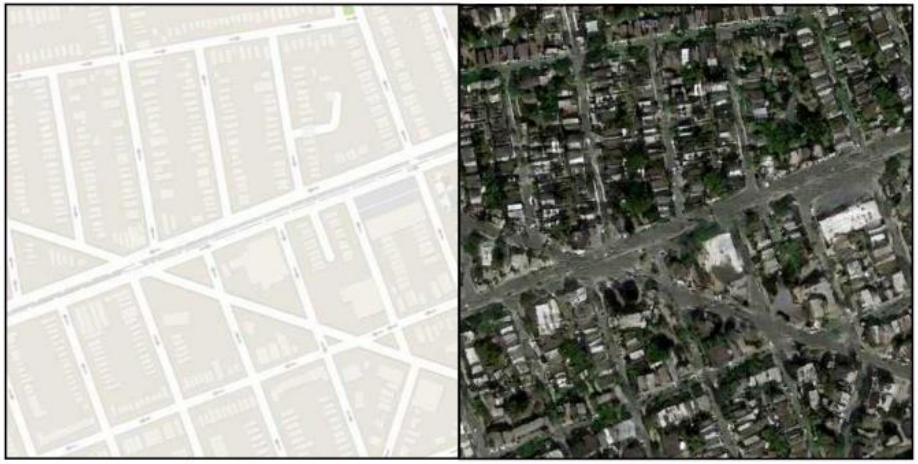
$$\mathcal{L}_{L1}(G) = \mathbb{E}_{x,y,z}[\|y - G(x,z)\|_1]$$

$$G^* = \arg\min_G \max_D \mathcal{L}_{cGAN}(G,D) + \lambda \mathcal{L}_{L1}(G)$$

Noise input z is not used as the algorithm simply ignores it

Results

Map to aerial photo



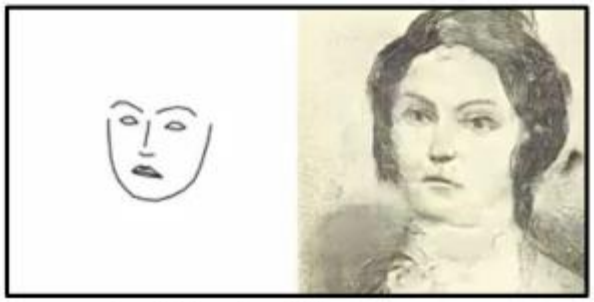
input output

Aerial photo to map



input output

Sketch→ Portrait



by Mario Klingemann

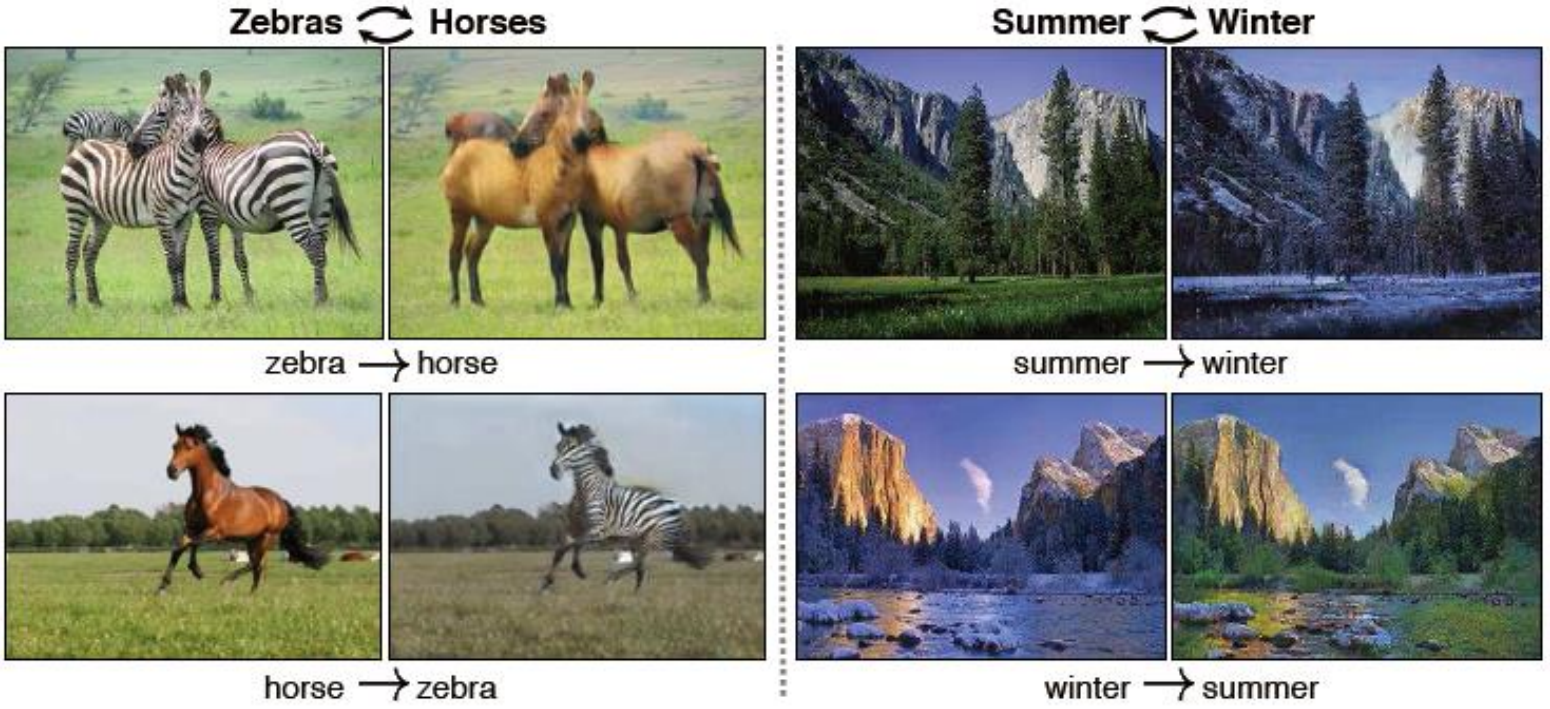
Depth→Streetview



by Jasper van Loenen

How about unpaired images?

CycleGAN

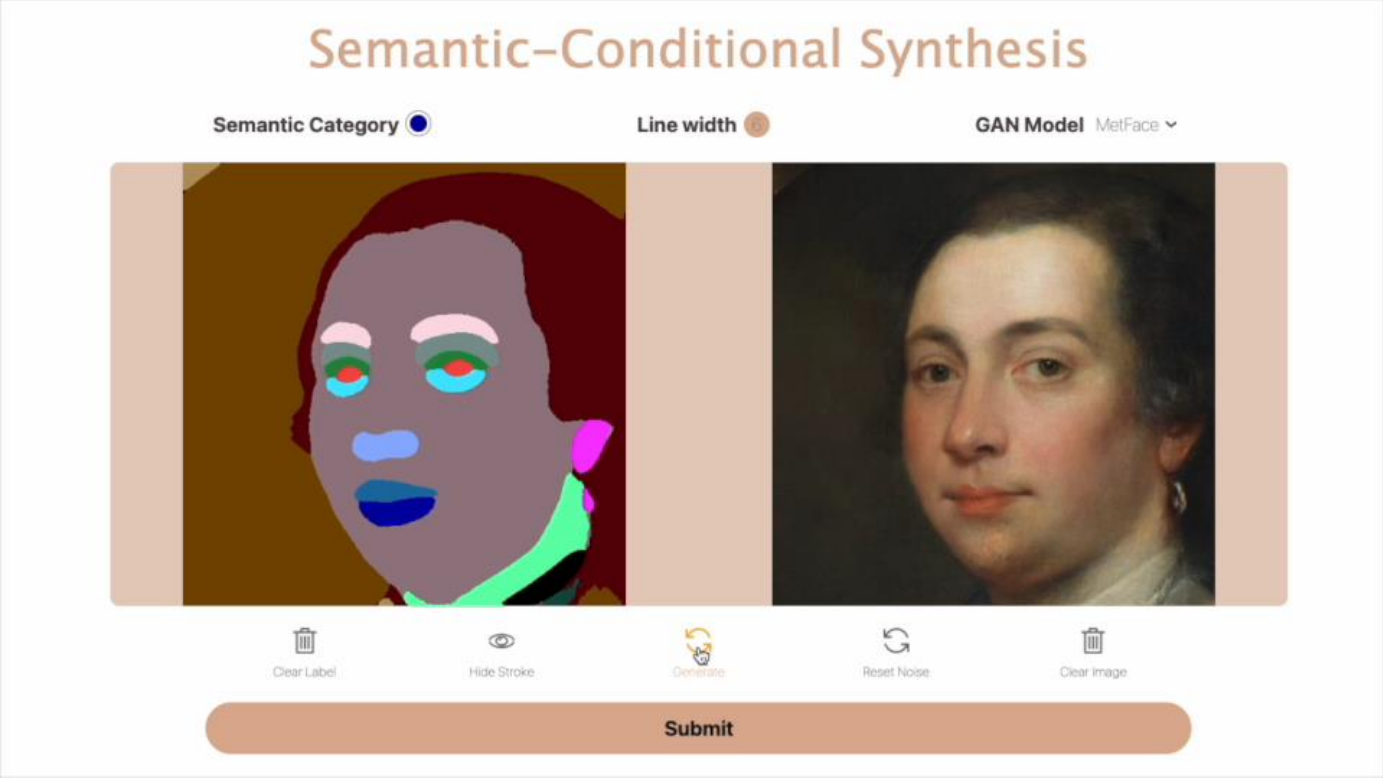


Please read: Zhu, Park, Isola, Efros, ICCV 2017

How to generate images interactively?

KLiSH

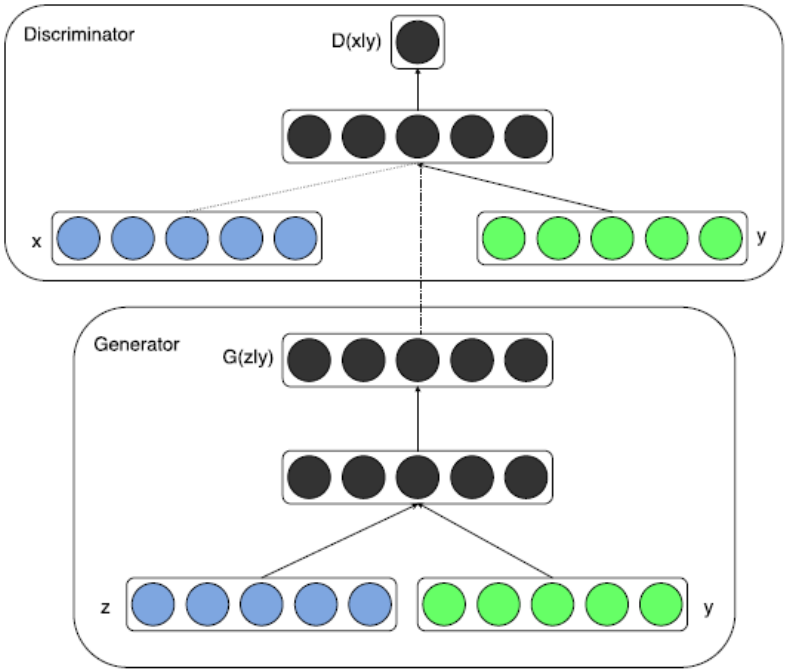
GANs' features are linearly separable and form semantic clusters



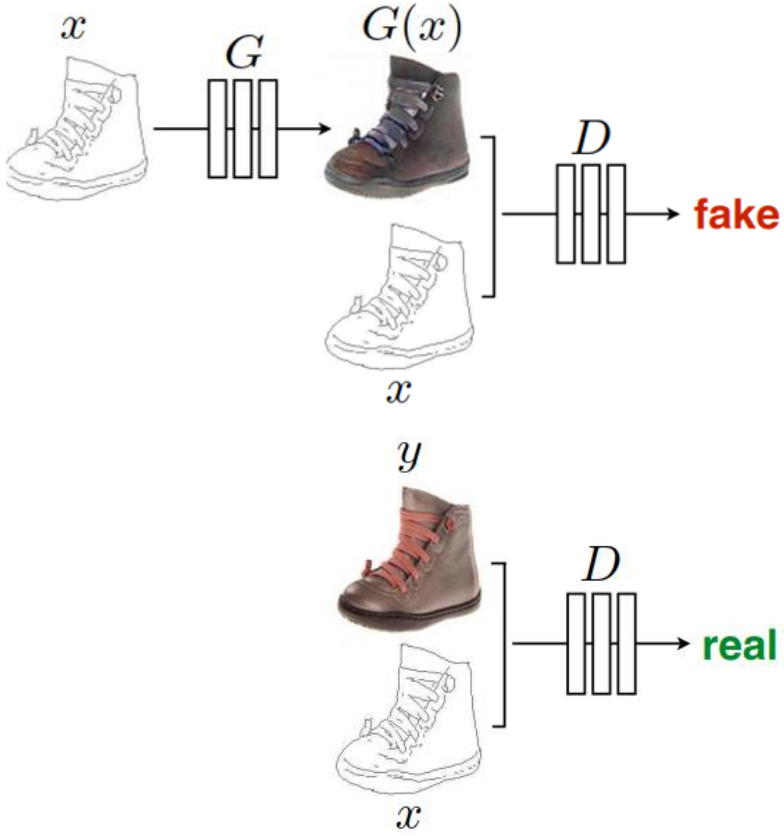
Please read: Xu, Zhang, Hu, IEEE TPAMI 2023

Summary of Part III

Conditional GAN



Pix2Pix GAN



Outline

I. The basic GAN
II. GANs for high-quality results
III. GANs for controllable generation
IV. Summary

Summary of this lecture

Knowledge

I. The basic GAN

$$\min_G \max_D V(D,G) = \mathbb{E}_{\mathbf{x}\sim p_{\mathrm{d}}(\boldsymbol{x})}[\log D(\boldsymbol{x})] + \mathbb{E}_{\mathbf{z}\sim p_z(\boldsymbol{z})}\left[\log\left(1 - D\left(G(\boldsymbol{z})\right)\right)\right]$$

II. GANs for high-quality results

DCGAN

StyleGAN

III. GANs for controllable generation

Conditional GAN

Pix2Pix GAN

Summary of this lecture

Capability and value

Min KL-divergence between $P_{\text{data}}(\mathbf{x})$ and $P_{\text{gen}}(\mathbf{x})$ ⇒ Max likelihood $P_{\text{gen}}(\mathbf{x})$

Go back to the original problem:

Design a generator (G) to generate images which are *very much like* the images in the training set!

Use a well-trained neural network to judge

References

- Goodfellow, Pouget-Abadiey, Mirza, et al. (2014)
  Generative Adversarial Nets
  NeurIPS
- Mirza, Osindero (2014)
  Conditional Generative Adversarial Nets
  arXiv:1411.1784v1
- Radford, Metz, Chintala (2016)
  Unsupervised representation learning with deep convolutional generative adversarial networks
  arXiv:1511.06434v2
- Karras, Laine, Aila (2019)
  A style-based generator architecture for generative adversarial networks.
  CVPR
- Isola, Zhu, Zhou, Efros (2017)
  Image-to-image translation with conditional adversarial networks
  CVPR

Recommended reading

- Zhu, Park, Isola, Efros (2017)

  Unpaired image-to-image translation using cycle-consistent adversarial networks

  ICCV
- Jianjin Xu, Zhaoxiang Zhang, Xiaolin Hu (2023)

  Extracting semantic knowledge from GANs with unsupervised learning

  IEEE TPAMI